Harris Hardiman-Mostow

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Summary — Final-year mathematics Ph.D. student and NSF Graduate Research Fellow with extensive machine learning research experience in academia and industry, spearheading complex projects with publications in respected venues. Expert in deep learning, graph learning, and active learning theory, algorithms, and applications. Proven leader in interdisciplinary teams and go-to person for scientific writing and conveying technical information to non-technical audiences.

Education

University of California, Los Angeles (UCLA) — **Expected, June 2026**
Ph.D. in Mathematics. Advisor: Andrea Bertozzi. — *GPA: 3.93/4.0*

- Awarded a 3-year National Science Foundation Graduate Research Fellowship ($149,000), the USA's oldest fellowship for STEM graduate students. Acceptance rate of approximately 15%. Alumni include 42 Nobel laureates.
- Won a 3-year National Defense Science and Engineering Graduate Fellowship ($136,000, declined), a highly competitive graduate STEM fellowship. Acceptance rate of approximately 7%.
- Received an NSF MENTOR Fellowship ($34,000), a training grant for early-career Ph.D. students in machine learning.
- Teaching Assistant for Linear Algebra, Differential Equations, and Integration and Infinite Series.

University of California, Los Angeles (UCLA) — **2023**
M.A. in Mathematics — *GPA: 3.92/4.0*

Tufts University — **2021**
B.S. in Mathematics, B.S. in Mechanical Engineering, Summa Cum Laude — *GPA: 3.96/4.0*

- Frederick Melvin Ellis Prize, for "marked athletic versatility, academic achievement, and effective leadership."
- Ralph S. Kaye Memorial Prize, awarded to the top mathematics student.
- Elected by 40 teammates as Captain of the Men's Varsity Rowing Team. First Team All-Conference and All-Academic.
- Teaching Assistant for Introduction to Computing.

Experience

University of California, Los Angeles (UCLA) — **2022 – Present**
Graduate Researcher, Fellow — *Los Angeles, CA*

- Leading research projects in deep learning, graph-based learning, and active learning, with applications to imaging, remote sensing and adversarial robustness. 5 (co)-first author publications currently accepted or submitted.

NASA Jet Propulsion Laboratory (JPL) — **07/2024 – 09/2024**
Graduate Machine Learning Research Intern — *La Cañada Flintridge, CA*

- Led effort to research, develop, train, and validate a novel transformer-based deep learning model to map natural disaster damage extents using synthetic aperture radar (SAR). Now the primary algorithm in JPL's SAR Disturbance Product.
- Collaborated with radar engineers and earth scientists to maximize efficacy of model and ensure real-world impact.
- Presented to groups within JPL and at the Science Understanding through Data Science (SUDS) Conference at Caltech.
- Featured on the NASA Disasters website in response to 2025 wildfires in Southern California.

The MITRE Corporation — **06/2021 – 07/2021**
Graduate Data Science Intern — *Bedford, MA*

- Researched and implemented unsupervised algorithms for multivariate online drift detection in time series data.

Publications

- **H. Hardiman-Mostow**, J. Mauro, A. Weihs, A.L. Bertozzi. Topology-Aware Active Learning on Graphs. Submitted to *Pattern Recognition*. arXiv:2510.25892.
- **H. Hardiman-Mostow**, C. Marshak, A. Handwerger. Deep Self-Supervised Global Disturbance Mapping with Sentinel-1 OPERA RTC Synthetic Aperture Radar. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*.
- J. Brown*, B. Chen*, **H. Hardiman-Mostow***, J. Calder, and A.L. Bertozzi. GLL: A Differentiable Graph Learning Layer for Neural Networks. In Revision, *Journal of Machine Learning Research*. arXiv:2412.08016.
- J. Brown*, B. Chen*, **H. Hardiman-Mostow***, A. Weihs*, A.L. Bertozzi and J. Chanussot. Material identification in complex environments: neural network approaches to hyperspectral image analysis. *IEEE WHISPERS*, 2023.
- J. Enwright*, **H. Hardiman-Mostow***, J. Calder, and A.L. Bertozzi. Deep semi-supervised label propagation with applications to SAR image classification. *SPIE Conference on Defense and Commercial Sensing*, 2023.

Skills

- **Expertise**: Deep learning, graph-based learning, active learning, self-supervised and semi-supervised learning, remote sensing, numerical analysis and numerical linear algebra, time-series data, data visualization, technical writing.
- **Programming**: Python (including numpy, PyTorch, sklearn, matplotlib, numba), MATLAB.
- **Software**: LaTeX, Git and GitHub, Weights and Biases (wandb), High Performance Computing (HPC), Microsoft Office.

Interests

Running, cycling, watching movies, cooking, food and wine, poker, watching basketball & football.